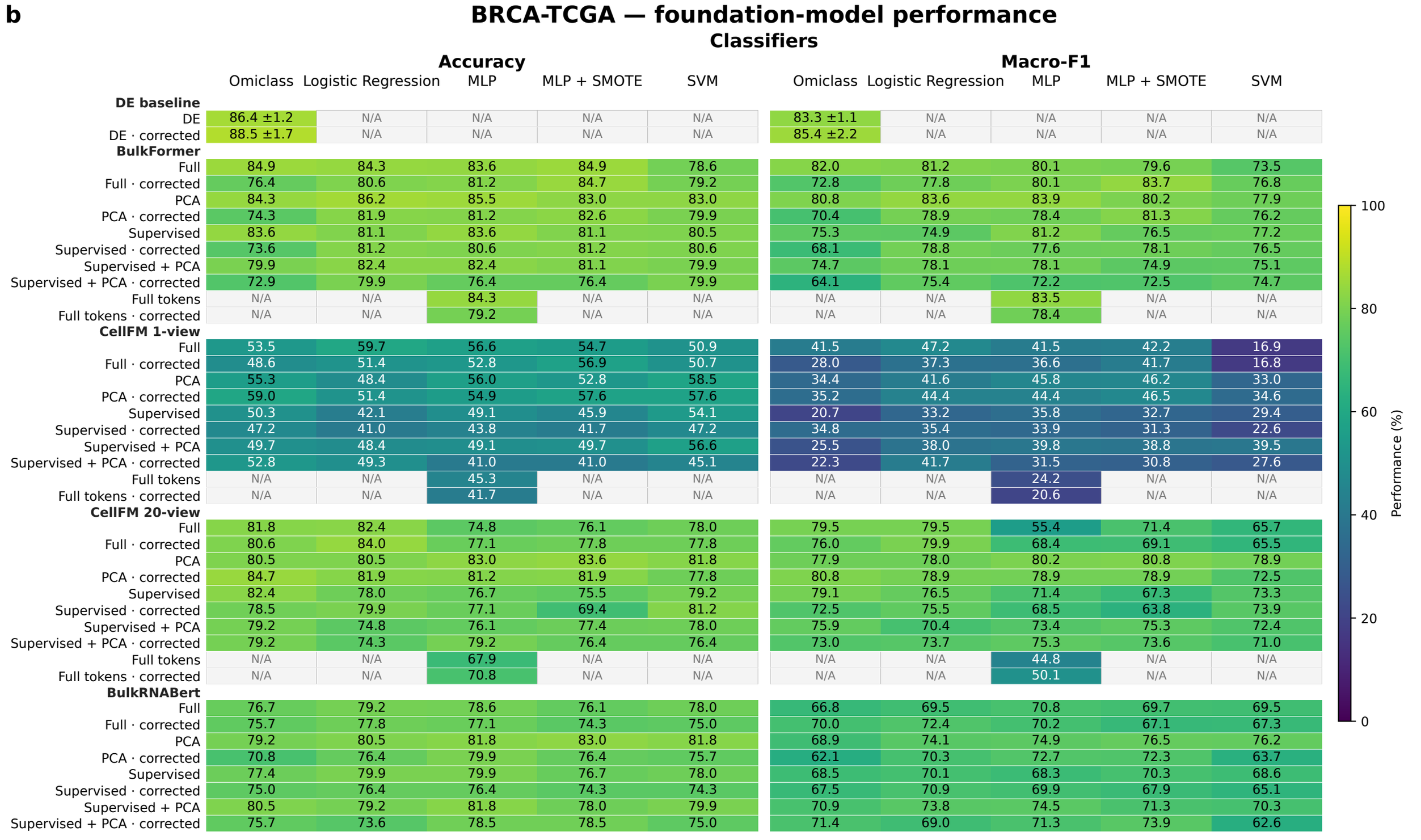
b
BRCA-TCGA — foundation-model performance
Classifiers
Accuracy
Macro-F1
Omiclass
Logistic Regression
MLP
MLP + SMOTE
SVM
DE baseline
DE 86.4 ±1.2 N/A N/A N/A N/A 83.3 ±1.1 N/A N/A N/A N/A
DE · corrected 88.5 ±1.7 N/A N/A N/A N/A 85.4 ±2.2 N/A N/A N/A N/A
BulkFormer
Full 84.9 84.3 83.6 84.9 78.6 82.0 81.2 80.1 79.6 73.5
Full · corrected 76.4 80.6 81.2 84.7 79.2 72.8 77.8 80.1 83.7 76.8
PCA 84.3 86.2 85.5 83.0 83.0 80.8 83.6 83.9 80.2 77.9
PCA · corrected 74.3 81.9 81.2 82.6 79.9 70.4 78.9 78.4 81.3 76.2
Supervised 83.6 81.1 83.6 81.1 80.5 75.3 74.9 81.2 76.5 77.2
Supervised · corrected 73.6 81.2 80.6 81.2 80.6 68.1 78.8 77.6 78.1 76.5
Supervised + PCA 79.9 82.4 82.4 81.1 79.9 74.7 78.1 78.1 74.9 75.1
Supervised + PCA · corrected 72.9 79.9 76.4 76.4 79.9 64.1 75.4 72.2 72.5 74.7
Full tokens N/A N/A 84.3 N/A N/A N/A N/A 83.5 N/A N/A
Full tokens · corrected N/A N/A 79.2 N/A N/A N/A N/A 78.4 N/A N/A
CellFM 1-view
Full 53.5 59.7 56.6 54.7 50.9 41.5 47.2 41.5 42.2 16.9
Full · corrected 48.6 51.4 52.8 56.9 50.7 28.0 37.3 36.6 41.7 16.8
PCA 55.3 48.4 56.0 52.8 58.5 34.4 41.6 45.8 46.2 33.0
PCA · corrected 59.0 51.4 54.9 57.6 57.6 35.2 44.4 44.4 46.5 34.6
Supervised 50.3 42.1 49.1 45.9 54.1 20.7 33.2 35.8 32.7 29.4
Supervised · corrected 47.2 41.0 43.8 41.7 47.2 34.8 35.4 33.9 31.3 22.6
Supervised + PCA 49.7 48.4 49.1 49.7 56.6 25.5 38.0 39.8 38.8 39.5
Supervised + PCA · corrected 52.8 49.3 41.0 41.0 45.1 22.3 41.7 31.5 30.8 27.6
Full tokens N/A N/A 45.3 N/A N/A N/A N/A 24.2 N/A N/A
Full tokens · corrected N/A N/A 41.7 N/A N/A N/A N/A 20.6 N/A N/A
CellFM 20-view
Full 81.8 82.4 74.8 76.1 78.0 79.5 79.5 55.4 71.4 65.7
Full · corrected 80.6 84.0 77.1 77.8 77.8 76.0 79.9 68.4 69.1 65.5
PCA 80.5 80.5 83.0 83.6 81.8 77.9 78.0 80.2 80.8 78.9
PCA · corrected 84.7 81.9 81.2 81.9 77.8 80.8 78.9 78.9 78.9 72.5
Supervised 82.4 78.0 76.7 75.5 79.2 79.1 76.5 71.4 67.3 73.3
Supervised · corrected 78.5 79.9 77.1 69.4 81.2 72.5 75.5 68.5 63.8 73.9
Supervised + PCA 79.2 74.8 76.1 77.4 78.0 75.9 70.4 73.4 75.3 72.4
Supervised + PCA · corrected 79.2 74.3 79.2 76.4 76.4 73.0 73.7 75.3 73.6 71.0
Full tokens N/A N/A 67.9 N/A N/A N/A N/A 44.8 N/A N/A
Full tokens · corrected N/A N/A 70.8 N/A N/A N/A N/A 50.1 N/A N/A
BulkRNABert
Full 76.7 79.2 78.6 76.1 78.0 66.8 69.5 70.8 69.7 69.5
Full · corrected 75.7 77.8 77.1 74.3 75.0 70.0 72.4 70.2 67.1 67.3
PCA 79.2 80.5 81.8 83.0 81.8 68.9 74.1 74.9 76.5 76.2
PCA · corrected 70.8 76.4 79.9 76.4 75.7 62.1 70.3 72.7 72.3 63.7
Supervised 77.4 79.9 79.9 76.7 78.0 68.5 70.1 68.3 70.3 68.6
Supervised · corrected 75.0 76.4 76.4 74.3 74.3 67.5 70.9 69.9 67.9 65.1
Supervised + PCA 80.5 79.2 81.8 78.0 79.9 70.9 73.8 74.5 71.3 70.3
Supervised + PCA · corrected 75.7 73.6 78.5 78.5 75.0 71.4 69.0 71.3 73.9 62.6
Performance (%)
100
80
60
40
20
0